# DIFFERENTIAL COHOMOLOGY SEMINAR 2

TALK BY MATTHIAS FRERICHS

In this lecture we want to learn the basics of $\infty$-category theory. For the $\infty$-categorical background, we broadly follow [Gro10] and a little [Lur09]. This talk assumes familiarity with the language of model categories, see [Hir03].

## 1. Basics on $(\infty, 1)$-categories

There are a pletora of models for $\infty$-categories, here we focus on two. We begin by recalling some definitions about simplicial sets.

**Definition 1.1.** Given a natural number $n$, denote by $\langle n \rangle$ the partially ordered set $\{0, \cdots, n\}$. The *simplex category* $\mathbf{\Delta}$ is the category of finite linerly ordered sets of the form $\langle n \rangle$, for any $n$, and monotone functions.

We single out two classes of functions in $\mathbf{\Delta}$:

**Definition 1.2.** Given $0 \leq i \leq n$, the *$i$-coface map* is the unique injective monotone function $\delta_n^i : \langle n-1 \rangle \to \langle n \rangle$ such that $i$ does not belong in the image of $\delta_n^i$. The *$i$-codegeneracy map* is the unique surjective monotone function $\sigma_n^i : \langle n+1 \rangle \to \langle n \rangle$ such that $i$ and $i+1$ have the same image.

**Theorem 1.3** ([GJ99, I.4])**.** *As a category, $\mathbf{\Delta}$ is generated from the face and degeneracy maps subject to the* simplicial identities.

**Definition 1.4.** A *simplicial set* is a contravariant functor $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$. The category of simplicial sets is denoted by $\mathrm{s}\mathcal{S}\mathrm{et}$. An element of $X_n = X(\langle n \rangle)$ is called a *$n$-simplex*. The function $X_n \to X_{n-1}$, resp $X_n \to X_{n+1}$, induced by $\delta_n^i$, resp. $X(\sigma_n^i)$, is called the *$i$-face map*, resp. *$i$-degeneracy map*, and denoted $d_i^n$, resp. $s_i^n$.

A simplicial set can be depicted as follows:

$$X_0 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow}\; X_1 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow}\; X_2 \;\substack{\longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow\\ \longleftarrow}\; \cdots \tag{1.5}$$

**Definition 1.6.** Given $0 \leq i \leq n$, the *$i$-horn* is the simplicial sub-set $\Lambda_i^n \subseteq \Delta^n$ defined as follows: A monotone map $f : \langle k \rangle \to \langle n \rangle$ is a $k$-simplex of $\Lambda_i^n$ if there is $j \neq i$ that does not belong in the image of $f$. A horn $\Lambda_i^n$ is *inner* if $0 < i < n$.

**Definition 1.7.** A simplicial set $X$ is called a *$\infty$-category* if, for every $n$ and $0 < i < n$, every map of simplicial sets $\Lambda_i^n \to X$ admits an extension to a map $\Delta^n \to X$. If the condition is verified for every $n$ and $0 \leq i \leq n$, $X$ is called a *$\infty$-groupoid*.

*Remark* 1.8. A map of simplicial sets $f : X \to Y$ is called a *Kan fibration* if it has the *horn lifting property*: For every $n$ and every $0 \leq i \leq n$, every diagram

$$\begin{tikzcd} \Lambda_n^i \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd} \tag{1.9}$$

admits a dashed arrow making both triangles commute. A map $f$ is a *inner Kan fibration* if the horn lifting property holds only for inner horns, i. e. $0 < i < n$. A simplicial set $X$ is a $\infty$-groupoid, resp. $\infty$-category, if and only if the terminal map $X \to \Delta^0$ is a Kan fibration, resp. inner Kan fibration.

**Definition 1.10.** Given a category $\mathcal{C}$, its *nerve* is the simplicial set $\mathrm{N}\mathcal{C}$ defined as follows: For $n \geq 1$, a $n$-simplex of $\mathrm{N}\mathcal{C}$ consists of a $n$-tuple of composable arrows. The objects of $\mathcal{C}$ are the 0-simplices.

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**Proposition 1.11.** *$N\mathcal{C}$ is a $\infty$-category.*

Viewed as a functor from categories to $\infty$-categories, the nerve functor admits a left adjoint called the *homotopy category* functor Ho. We will sketch the construction of $\mathrm{Ho}X$, refering to [Lan21] and [Gro10] for more details.

*Remark* 1.12. Given a 1-simplex $f$, we write $f : x \to y$ to mean that $d_1^1(f) = x$ and $d_0^1(f) = y$. The 1-simplex $s_0^0(x)$ is written as

$$x = x \tag{1.13}$$

**Definition 1.14.** Given a simplicial set $X$, two 1-simplices $f, g : x \to y$ are *homotopic* if there is a 2-simplex of the form:

$$\begin{tikzcd} & y \arrow[dr, equal] & \\ x \arrow[ur, "f"] \arrow[rr, "g"'] & & y \end{tikzcd}$$

Given 0-simplices $x, y$, denote by $\hom_{\mathrm{Ho}(X)}(x, y)$ the set of 1-simplices $x \to y$ modulo the equivalence relation generated by being homotopic. Given a 1-simplex $f : x \to y$, its homotopy class in $\hom_{\mathrm{Ho}X}(x, y)$ is denoted $[f]$.

**Lemma 1.15** ([Lur26e])**.** *Let $X$ be an $\infty$-category, then:*

1. *For every pair $x, y$ of 0-simplices, being homotopic is an equivalence relation on the set of 1-simplices $x \to y$.*
2. *For every pair $f : x \to y$ and $g : y \to z$, there is a 2-simplex of the form*

$$\begin{tikzcd} & y \arrow[dr, "g"] & \\ x \arrow[ur, "f"] \arrow[rr, "h"'] & & z \end{tikzcd} \tag{1.16}$$

3. *The homotopy class of $h$ in Equation (1.16) only depends on the homotopy classes of $f$ and $g$.*
4. *For every triple $x, y, z$ of 0-simplices, there is a composition law*

$$\hom_{\mathrm{Ho}X}(x, y) \times \hom_{\mathrm{Ho}X}(y, z) \xrightarrow{\circ_{x,y,z}} \hom_{\mathrm{Ho}X}(x, z) \tag{1.17}$$

   *defined as follows: Given $f : x \to y, g : y \to z$, let $h$ be a 1-simplex as in Equation (1.16), then $[g] \circ [f] := [h]$.*
5. *The composition laws $\circ_{x,y,z}$ are associative.*
6. *Given a 0-simplex $x$, the left homotopy class of $s_0^0(x)$ acts as the left and right identity with respect to the composition laws above.*

**Definition 1.18.** Let $X$ be a $\infty$-category, the *homotopy category*, denoted $\mathrm{Ho}X$, is the category defined as follows:

1. The set of objects is $X_0$.
2. Given a pair of objects $x, y$, let $\hom_{\mathrm{Ho}X}(x, y)$ be the set of morphisms $x \to y$.
3. Take $\circ_{x,y,z}$ from Equation (1.15) as composition laws.

**Definition 1.19.** Let $X$ be a $\infty$-category, a 1-simplex $f : x \to y$ is an *equivalence* if $[f]$ is an isomorphism in $\mathrm{Ho}X$.

The following theorem shows that $\infty$-groupoids have a relation to $\infty$-categories similar to that of ordinary groupoids to ordinary categories.

**Theorem 1.20** ([Lur26b])**.** *A quasi-category $X$ is a $\infty$-groupoid if and only if $\mathrm{Ho}X$ is a groupoid.*

$\infty$-groupoids naturally form a full sub-category of $s\mathcal{S}\mathrm{et}$. To construct the $\infty$-category of $\infty$-groupoids, we use the simplicial nerve of Lurie.

**Definition 1.21.** Denote by $s\mathcal{C}\mathrm{at}$ the category of simplicially enriched categories. A simplicially enriched category is called *locally fibrant* if its hom-simplicial sets are $\infty$-groupoids.

Recall that $s\mathcal{S}\mathrm{et}$ is self-enriched as follows:

**Definition 1.22.** Given two simplicial sets $X, Y$, the *hom-simplicial set* $\hom^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(X, Y)$ is defined as follows: A $n$-simplex consists of a map of simplicial sets $\Delta^n \times X \to Y$.

**Lemma 1.23.** *If $X$ is a $\infty$-groupoid, then $\hom^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(B, X)$ is a $\infty$-groupoid, for every simplicial set $B$.*

*Proof.* Corollary of [Lur26a] by taking $f$ the terminal map $X \to \Delta^0$ and $i$ the initial map $\emptyset \to B$. □

**Definition 1.24.** Let $(P, \leq)$ be a partially ordered set, denote by $\mathrm{Path}(P)$ the following simplicial category:

(1) Objects are the elements of $P$.
(2) Given $x, y \in P$, let $[x, y] := \{z \in P | x \leq z \leq y\}$ and $P_{x,y}$ the set of sub-sets $S \subseteq [x, y]$ that are finite, linearly ordered, and contain $\{x, y\}$. We order $P_{x,y}$ by reverse inclusion ($S \leq T$ if $S \supseteq T$). Finally, let $\hom_{\mathrm{Path}(P)}(x, y)$ be the nerve of $(P_{x,y}, \supseteq)$.
(3) Given $x, y, z \in P$, composition

$$\hom_{\mathrm{Path}(P)}(x, y) \times \hom_{\mathrm{Path}(P)}(y, z) \longrightarrow \hom_{\mathrm{Path}(P)}(x, z) \tag{1.25}$$

is the nerve of the union map $\cup : P_{x,y} \times P_{y,z} \to P_{x,z}$.

Given a monotone map $f : P \to Q$, let $\mathrm{Path}(f) : \mathrm{Path}(P) \to \mathrm{Path}(Q)$ denote the following functor:

(1) On objects, it is simply $f$.
(2) Given $x, y \in P$, define $\mathrm{Path}(f)$ as the nerve of the functor $f_* : P_{x,y} \to Q_{f(x),f(y)}$ given by taking direct images (i. e. $f_*(S) = f(S)$).

Restricting to $\mathbf{\Delta}$, we obtain a cosimplicial simplicial category $\mathrm{Path} : \mathbf{\Delta} \to \mathrm{s}\mathcal{C}\mathrm{at}$.

**Definition 1.26.** Given a simplicial category $\mathcal{C}$, its *simplicial nerve* is the simplicial set $\mathrm{N}_{\Delta}\mathcal{C}$ defined as follows: A $n$-simplex of $\mathrm{N}_{\Delta}\mathcal{C}$ consists of a functor $\mathrm{Path}_n := \mathrm{Path}(\langle n \rangle) \to \mathcal{C}$.

*Remark* 1.27. $\mathrm{Path}_0$ and $\mathrm{Path}_1$ are the categories $\langle 0 \rangle$ and $\langle 1 \rangle$ viewed as simplicially enriched categories with discrete hom-simplicial sets, therefore a functor $\mathrm{Path}_0 \to \mathcal{C}$, resp. $\mathrm{Path}_1 \to \mathcal{C}$, is determined by a single object, resp. 1-morphism. $\mathrm{Path}_2$ admits the following description:

(1) Its set of objects is $\langle 2 \rangle$.
(2) $P_{0,1}$ and $P_{1,2}$ are both singletons, hence $\hom_{\mathrm{Path}_2}(0, 1) \cong \hom_{\mathrm{Path}_2}(1, 2) \cong \Delta^0$.
(3) $P_{0,2} = \{[0, 2] \leq \{0, 2\}\}$ (recall that $P_{i,j}$ is ordered by reversed inclusion), so $\hom_{\mathrm{Path}_2}(0, 2) \cong \Delta^1$. Moreover, $[0, 2]$ is the composition $\{1, 2\} \circ \{0, 1\}$.

We can conclude that a functor $\mathrm{Path}_2 \to \mathcal{C}$ is determined by the following data: A triple of 1-morphisms $f : x \to y, g : y \to z$, and $h : x \to y$, together with a 1-simplex $g \circ f \Rightarrow h$ in $\hom_{\mathcal{C}}(x, z)$.

**Definition 1.28.** Denote also by Path the functor $\mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{s}\mathcal{C}\mathrm{at}$ obtained by left Kan extension of $\mathrm{Path} : \mathbf{\Delta} \to \mathrm{s}\mathcal{C}\mathrm{at}$.

**Theorem 1.29.** *The functor* Path *is left adjoint to* $\mathrm{N}_{\Delta}$ *and form a Quillen equivalence between the* Dwyer-Kan-Bergner *model structure on* $\mathrm{s}\mathcal{C}\mathrm{at}$ *and the* Joyal *model structure on* $\mathrm{s}\mathcal{S}\mathrm{et}$. *The fibrant objects in the* $\mathrm{s}\mathcal{S}\mathrm{et}_{\mathrm{Joyal}}$, *resp.* $\mathrm{s}\mathcal{C}\mathrm{at}_{\mathrm{DKB}}$, *are $\infty$-categories, resp. locally fibrant categories. In particular,* $\mathrm{N}_{\Delta}\mathcal{C}$ *is a $\infty$-category if $\mathcal{C}$ is locally fibrant.*

By Equation (1.23), we know that the full sub-category $\infty\mathcal{G}\mathrm{rpd} \subseteq \mathrm{s}\mathcal{S}\mathrm{et}$ of $\infty$-groupoids is self-enriched, therefore we give the following definition.

**Definition 1.30.** Denote by $\mathcal{S} := \mathrm{N}_{\Delta}(\infty\mathcal{G}\mathrm{rpd})$ the *$\infty$-category of $\infty$-groupoids.*

*Remark* 1.31. In view of [Lur26c], we will also use the term *space* to mean $\infty$-groupoid.

## 2. Basic constructions

Recall that $\mathrm{s}\mathcal{S}\mathrm{et}$ and $\infty\mathcal{G}\mathrm{rpd}$ are self-enriched. Similar to Equation (1.23), we have the following:

**Lemma 2.1** ([Lur26d])**.** *If $X$ is a $\infty$-category, then $\hom^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(B, X)$ is a $\infty$-category, for every simplicial set $B$.*

**Definition 2.2.** Given a $\infty$-category $X$ and simplicial set $B$, denote by $\mathcal{F}\mathrm{un}(B, X)$ the *$\infty$-category of functors* $\hom^{\Delta}_{\mathrm{s}\mathcal{S}\mathrm{et}}(B, X)$.

**Definition 2.3.** Denote by op the following endo-functor of $\mathbf{\Delta}$:

(1) On objects, $\mathrm{op}(\langle n\rangle) = \langle n\rangle$.
(2) Given $f : \langle k\rangle \to \langle n\rangle$, let $\mathrm{op}(f) : \langle k\rangle \to \langle n\rangle$ be the function $\mathrm{op}(f)(i) = n - f(k-i)$.

Given a simplicial set $X$, denote by $X^{\mathrm{op}}$ the pre-composition of $X$ with op.

**Lemma 2.4.** *If $X$ is a $\infty$-category, $X^{\mathrm{op}}$ is also an $\infty$-category.*

Slice $\infty$-categories?

**Definition 2.5.** Given a $\infty$-category $X$, we call $X^{\mathrm{op}}$ the *opposite $\infty$-category.*

## 3. Accessible and Presentable Categories

In general, a limit, resp. colimit, preserving functor need not have a left, resp. right, adjoint. Here we wish to introduce a rather large class of $\infty$-categories for which the previous statement holds. We begin by recalling the definition of locally presentable 1-categories. Let $\kappa$ denote a regular cardinal.

**Definition 3.1.** A category $\mathcal{I}$ is *$\kappa$-filtered* if, for every $\mathcal{J}$ with $< \kappa$ many morphisms, every diagram $\mathcal{J} \to \mathcal{I}$ has a cocone. A functor $F : \mathcal{C} \to \mathcal{D}$ is *$\kappa$-accessible* if it preserves $\kappa$-filtered colimits. Given a category $\mathcal{C}$, an object $X$ is *$\kappa$-compact* if $\hom_{\mathcal{C}}(X, -) : \mathcal{C} \to \mathcal{S}\mathrm{et}$ is $\kappa$-accessible.

**Definition 3.2.** A category $\mathcal{C}$ is *$\kappa$-accessible* if there exists a set $S \subseteq \mathcal{C}_0$ of $\kappa$-compact objects that generate $\mathcal{C}$ under $\kappa$-filtered colimits. A category is *accessible* if it is $\kappa$-accessible, for some regular cardinal $\kappa$.

**Definition 3.3.** A category $\mathcal{C}$ that is accessible and cocomplete is called *locally presentable.*

**Theorem 3.4.** *Let $\mathcal{C}$ be a category, then $\mathcal{C}$ is locally presentable if and only if there exists a small category $S$ such that the induced functor $\mathcal{C} \to \mathcal{P}(S)$ is a fully faithful, accessible right adjoint.*

**Theorem 3.5.** *Let $\mathcal{C}, \mathcal{D}$ be locally presentable categories. A functor $F\colon \mathcal{C} \to \mathcal{D}$ is a left, resp. right, adjoint if and only if it preserves colimits, resp. it preserves limits and is accessible.*

**Definition 3.6.** Let $X$ be a $\infty$-category. The *$\infty$-category of pre-sheaves of spaces* is defined as $\mathcal{P}(X) := \hom_{\mathrm{sSet}}(X^{\mathrm{op}}, \mathcal{S})$.

**Theorem 3.7** (Yoneda)**.** *Given a $\infty$-category $X$, there is a fully faithful functor $y : X \to \mathcal{P}(X)$, called the* Yoneda embedding*, such that: Given a cocomplete $\infty$-category $Y$, pre-composition by $y$ induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{P}(X), Y) \longrightarrow \mathcal{F}\mathrm{un}(X, Y) \tag{3.8}$$

*where $\mathcal{F}\mathrm{un}^{\mathrm{L}}$ denotes the category of colimit preserving functors.*

The definition of accessible category transfers directly to the $\infty$-categorical setting.

**Theorem 3.9.** *A $\infty$-category $X$ is locally presentable (cocomplete and accessible) if and only if there is a small sub-$\infty$-category $S$ such that $X \to \mathcal{F}\mathrm{un}(S^{\mathrm{op}}, \mathcal{S})$ is a fully faithful, accessible right adjoint.*

*Remark* 3.10. In view of Equation (3.9), one can define a category to be *locally presentable* if it is the accessible right localization of a pre-sheaf category for some small $\infty$-category $S$. In particular, every pre-sheaf category is locally presentable.

**Theorem 3.11.** *Let $X, Y$ be presentable $\infty$-categories, then a functor $f : X \to Y$ is a left, resp. right, adjoint if and only if it preserves colimits, resp. it preserves limits and is accessible.*

## 4. Stable $\infty$-Categories and Spectra

We now use the $\infty$-categorical framework to study spectra. Here are some result that justify the study of spectra.

Examples of stable phenomena? Freudenthal's suspension theorem, Alexander duality of manifolds, etc.

**Definition 4.1.** Let $\mathcal{C}$ be an $\infty$-category, an object 0 that is both initial and terminal is called *zero object.* A category $\mathcal{C}$ with a zero object is called a *pointed category.*

**Example 4.2.** Let $1 \in \mathcal{C}$ be a terminal object, then the identity of 1 is the zero object in the slice category $\mathcal{C}_{1/}$ of objects under 1. In particular, the category $\mathcal{S}_* = \mathcal{S}_{*/}$ of pointed spaces is pointed.

**Proposition 4.3.** *Let $\mathcal{D}$ be a pointed $\infty$-category. Evaluation at the 0-sphere $S^0$ induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{S}_*, \mathcal{C}) \longrightarrow \mathcal{C} \tag{4.4}$$

We now introduce the notion of a triangle.

**Definition 4.5.** Let $\mathcal{C}$ be a pointed $\infty$-category. A *triangle* in $\mathcal{C}$ is a commutative diagram of the form

$$\begin{tikzcd} X \arrow[r] \arrow[d] & Y \arrow[d] \\ 0 \arrow[r] & Z \end{tikzcd} \tag{4.6}$$

A triangle is *exact*, resp. *coexact*, if it is a pullback, resp. pushout, square.

**Definition 4.7.** Let $\mathcal{C}$ be a pointed $\infty$-category. Denote by $\mathcal{C}^{\Sigma}$, resp. $\mathcal{C}^{\Omega}$, the full sub-category of $\mathcal{F}\mathrm{un}(\Delta^1 \times \Delta^1, \mathcal{C})$ of coexact, resp. exact, triangles of the form

$$\begin{tikzcd} X \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & Y \end{tikzcd}, \tag{4.8}$$

If $\mathcal{C}$ is finitely cocomplete, resp. complete, for every object $X$, resp. $Y$, there is a contractible space of coexact, resp. exact, triangles as Equation (4.8). In particular, $\mathcal{C} \simeq \mathcal{C}^{\Sigma}$ and $\mathcal{C} \simeq \mathcal{C}^{\Omega}$.

**Proposition 4.9.** *If $\mathcal{C}$ is a finitely complete and cocomplete pointed $\infty$-category, define the following functors Then the functors*

$$\Sigma : \mathcal{C} \longrightarrow \mathcal{C}^{\Sigma} \xrightarrow{\mathrm{ev}_{(1,1)}} \mathcal{C} \qquad\qquad \Omega : \mathcal{C} \longrightarrow \mathcal{C}^{\Omega} \xrightarrow{\mathrm{ev}_{(0,0)}} \mathcal{C} \tag{4.10}$$

*are adjoint ($\Sigma$ is left adjoint to $\Omega$).*

**Theorem 4.11.** *Let $\mathcal{C}$ be a finitely bicomplete pointed $\infty$-category. The following are equivalent:*

*(1) A triangle is exact if and only if it is coexact.*
*(2) $(\Sigma, \Omega)$ is an adjoint equivalence.*
*(3) A commutative square is a pullback if and only if it is a pushout.*

**Definition 4.12.** A finite bicomplete pointed $\infty$-category $\mathcal{C}$ satisfying any of the equivalent conditions in Equation (4.11) is called *stable*.

If $\mathbb{A}$ denotes a nice abelian category, there is a *derived $\infty$-category* of $\mathbb{A}$, denoted $\mathcal{D}(\mathbb{A})$ such that $\mathrm{Ho}\mathcal{D}(\mathbb{A}) = D(\mathbb{A})$ is the ordinary derived category of $\mathbb{A}$, i.e. the localization of chain complex at quasi-isomorphisms. From homological algebra, it is known that $D(\mathbb{A})$ underlies the structure of a triangulated category, which turns out to be the 1-categorical reflection of $\mathcal{D}(\mathbb{A})$ being stable.

**Proposition 4.13** ([Lur17, 3.11])**.** *If $\mathcal{C}$ is a stable $\infty$-category, then* $\mathrm{Ho}\mathcal{C}$ *has a canonical structure of a triangulated category.*

To construct the stabilization of a pointed $\infty$-category, there are several approaches, such as reduced excisive functors on $\mathcal{S}_*^{\mathrm{fin}}$, the category of pointed, finite spaces, see [Lur17, 1.4.2.8]. Here we consider the more explicit approach using spectrum objects.

**Definition 4.14.** Let $\mathcal{C}$ be a pointed $\infty$-category. A *pre-spectrum object in $C$* consists of a functor $E : \mathbb{Z} \times \mathbb{Z} \to \mathcal{C}$ such that $E(n,m) \simeq 0$, for all $n \neq m$. Denote by $\mathcal{PS}\mathrm{p}(\mathcal{C})$ the category of pre-spectrum objects. The functor $\Omega^{\infty - n} : \mathcal{PS}\mathrm{p}(\mathcal{C}) \to \mathcal{C}$ is defined as evaluation at $(n,n)$.

For every $n$, the diagram

$$\begin{tikzcd} E(n,n) \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & E(n+1,n+1) \end{tikzcd} \tag{4.15}$$

determines a pair of adjoint morphisms

$$\alpha_n : \Sigma E(n,n) \to E(n+1,n+1), \qquad \beta_n : E(n,n) \to \Omega E(n+1,n+1)$$

**Definition 4.16.** Let $\mathcal{C}$ be a pointed $\infty$-category. A *spectrum object in* $\mathcal{C}$ consists of a pre-spectrum object $E$ such that $\beta_n$ is an equivalence, for all $n$. Denote by $\mathcal{S}\mathrm{p}(\mathcal{C}) \subseteq \mathcal{P}\mathcal{S}\mathrm{p}(\mathcal{C})$ the full sub-category of spectrum objects.

**Theorem 4.17.** *Let $\mathcal{C}$ be a presentable pointed $\infty$-category, then $\Omega^{\infty-n} : \mathcal{S}\mathrm{p}(\mathcal{C}) \to \mathcal{C}$ admits a left adjoint $\Sigma^{\infty-n} : \mathcal{C} \to \mathcal{S}\mathrm{p}(\mathcal{C})$, for every $n$.*

In particular, $\Sigma^\infty : \mathcal{C} \to \mathcal{S}\mathrm{p}(\mathcal{C})$ has the following universal property.

**Theorem 4.18.** *Let $\mathcal{C}$ be a presentable pointed $\infty$-category. Given a stable $\infty$-category $\mathcal{D}$, pre-composition by $\Sigma^\infty$ induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{S}\mathrm{p}(\mathcal{C}), \mathcal{D}) \longrightarrow \mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{C}, \mathcal{D}) \tag{4.19}$$

*In particular, for $\mathcal{C} = \mathcal{S}_*$, evaluation at the* sphere spectrum $\mathbb{S} = \Sigma^\infty S^0$ *induces an equivalence*

$$\mathcal{F}\mathrm{un}^{\mathrm{L}}(\mathcal{S}\mathrm{p}(\mathcal{S}_*), \mathcal{D}) \longrightarrow \mathcal{D} \tag{4.20}$$

**Definition 4.21.** The *$\infty$-category of spectra* is the category of spectrum objects in pointed spaces.

## 5. Generalized Cohomology Theories

We shall now use the language of $\infty$-categories to reformulate the concept of generalized cohomology theory à-là Eilenberg-Steenrod. In this new context, we recall a representability theorem for cohomology theories by spectrum object.

*Remark* 5.1. Denote by $\mathcal{S}\mathrm{et}^{\mathbb{Z}}$ the category of $\mathbb{Z}$-indexes families of sets. Given an object $S$ and $n \in \mathbb{Z}$, denote by $\Sigma^n S$ the shifted family $(\Sigma^n S)_i = S_{i-n}$.

**Definition 5.2** ([Lur17, 1.4.1.6])**.** Let $\mathcal{C}$ be a finitely cocomplete pointed $\infty$-category, $\Sigma_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ the induced suspension functor. A *generalized cohomology theory* is a functor $H : \mathrm{Ho}\mathcal{C}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}^{\mathbb{Z}}$ together with a natural isomorphism $\partial : \Sigma H \to H\Sigma_{\mathcal{C}}$ such that:

- $H$ preserves arbitrary products. In particular, $H^n(0)$ is the one-point set. Given an object $X$, the unique morphism $X \to 0$ induces an element $* \simeq H^n(0) \to H^n(X)$, which we denote by 0.
- Given a coexact triangle $X' \to X \to X''$, if $\eta \in H^n(X)$ has image $0 \in H^n(X'')$, then it lies in the image of $H^n(X') \to H^n(X)$.

**Theorem 5.3** ([Lur17, 1.4.1.10])**.** *Let $\mathcal{C}$ be a nice $\infty$-category and $(H, \partial)$ a generalized cohomology theory, then, for every $n$, the functor $H^n$ is representable by an object $E(n)$.*

The natural isomorphism $\partial$ translates into an equivalence $E(n) \simeq \Omega E(n+1)$, which is then used to construct a spectrum object representing the cohomology theory $H^n$, see [Lur17, 1.4.1.11].

*Remark* 5.4. For $\mathcal{C} = \mathcal{S}_*$, the above definition of cohomology theory reduces to the classical Eilenberg-Steenrod definition. Since $\mathcal{S}_*$ is nice, we thus recover the classical *Brown representability theorem.*

## References


[GJ99] Paul G. Goerss and John F. Jardine. *Simplicial homotopy theory*, volume 174 of *Progress in Mathematics*. Birkhäuser Verlag, Basel, 1999.

[Gro10] Moritz Groth. A short course on $\infty$-categories. *arXiv preprint*, 2010. arXiv:1007.2925.

[Hir03] Philip S. Hirschhorn. *Model categories and their localizations*, volume 99 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2003.

[Lan21] Markus Land. *Introduction to $\infty$-categories*. Birkhäuser Cham, 2021.

[Lur09] Jacob Lurie. *Higher topos theory*, volume 170 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 2009.

[Lur17] Jacob Lurie. Higher algebra. Available online, September 2017.

[Lur26a] Jacob Lurie. Kerodon (Tag 00TK). https://kerodon.net/tag/00TK, 2026.

[Lur26b] Jacob Lurie. Kerodon (Tag 019D). https://kerodon.net/tag/019D, 2026.

[Lur26c] Jacob Lurie. Kerodon (Tag 01Z4). https://kerodon.net/tag/01Z4, 2026.

[Lur26d] Jacob Lurie. Kerodon (tg 0066). https://kerodon.net/tag/0066, 2026.

[Lur26e] Jacob Lurie. The Homotopy Category of an $\infty$-Category. https://kerodon.net/tag/0049, 2026.